

Friday, Mar 29

Discrete Marginal Effects

Consider a regression model with (without loss of generality) two explanatory variables, X_1 and X_2 . A *discrete marginal effect* is the change in the expected response when we change an explanatory variable.

For example, if we have a regression model where $E(Y)$ is a function of X_1 and X_2 , the discrete marginal effect of changing X_1 from x_b to x_a is

$$E(Y|X_1 = x_a, X_2 = x_2) - E(Y|X_1 = x_b, X_2 = x_2).$$

That is, the change in the expected response when X_1 is changed from x_b to x_a . (Note: When we talk about a change in the expected response or the “effect” of a change in an explanatory variable, we do not necessarily mean that this is a *causal* relationship.)

In a linear model a discrete marginal effect is basically what is done by **contrast**.

Example: Recall our model for the `whiteside` data. The function `margeff` in the `trtools` package will estimate a discrete marginal effect.

```
m <- lm(Gas ~ Insul + Temp + Insul:Temp, data = MASS::whiteside)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.8538	0.13596	50.409	7.997e-46
InsulAfter	-2.1300	0.18009	-11.827	2.316e-16
Temp	-0.3932	0.02249	-17.487	1.976e-23
InsulAfter:Temp	0.1153	0.03211	3.591	7.307e-04

The model is

$$E(Y_i) = \beta_0 + \beta_1 a_i + \beta_2 t + \beta_3 a_i t_i,$$

where Y_i is gas consumption,

$$a_i = \begin{cases} 1, & \text{if the } i\text{-th observation is after insulation,} \\ 0, & \text{otherwise.} \end{cases}$$

So the marginal effect of increasing temperature from $t_b = 2$ to $t_a = 7$ after insulation is

$$E(Y|a = 1, t = 7) - E(Y|a = 1, t = 2) = 5(\beta_2 + \beta_3).$$

Before insulation it is

$$E(Y|a = 0, t = 7) - E(Y|a = 0, t = 2) = 5\beta_2.$$

We can estimate this using the `lincon` or `contrast` functions.

```
library(trtools)
lincon(m, a = c(0,0,5,5)) # marginal effect after insulation
```

	estimate	se	lower	upper	tvalue	df	pvalue
(0,0,5,5),0	-1.39	0.1146	-1.62	-1.16	-12.12	52	8.936e-17

```
lincon(m, a = c(0,0,5,0)) # marginal effect after insulation
```

	estimate	se	lower	upper	tvalue	df	pvalue
(0,0,5,0),0	-1.966	0.1124	-2.192	-1.741	-17.49	52	1.976e-23

```
contrast(m, cnames = c("Before","After"),
  a = list(Temp = 7, Insul = c("Before","After")),
  b = list(Temp = 2, Insul = c("Before","After")))
```

	estimate	se	lower	upper	tvalue	df	pvalue
Before	-1.966	0.1124	-2.192	-1.741	-17.49	52	1.976e-23
After	-1.390	0.1146	-1.620	-1.160	-12.12	52	8.936e-17

The function `margeff` (also from the `trtools` package) is specifically designed to estimate marginal effects (and other things) and works similarly to `contrast`.

```
margeff(m, cnames = c("Before","After"),
  a = list(Temp = 7, Insul = c("Before","After")),
  b = list(Temp = 2, Insul = c("Before","After")))
```

	estimate	se	lower	upper	tvalue	df	pvalue
Before	-1.966	0.1124	-2.192	-1.741	-17.49	52	1.976e-23
After	-1.390	0.1146	-1.620	-1.160	-12.12	52	8.936e-17

We can also estimate the discrete marginal effect of adding insulation at different temperatures.

```
contrast(m, cnames = c("0C","5C","10C"),
  a = list(Temp = c(0,5,10), Insul = "After"),
  b = list(Temp = c(0,5,10), Insul = "Before"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
0C	-2.1300	0.18009	-2.491	-1.769	-11.827	52	2.316e-16
5C	-1.5535	0.08777	-1.730	-1.377	-17.699	52	1.155e-23
10C	-0.9769	0.18583	-1.350	-0.604	-5.257	52	2.784e-06

```
margeff(m, cnames = c("0C","5C","10C"),
  a = list(Temp = c(0,5,10), Insul = "After"),
  b = list(Temp = c(0,5,10), Insul = "Before"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
0C	-2.1300	0.18009	-2.491	-1.769	-11.827	52	2.316e-16
5C	-1.5535	0.08777	-1.730	-1.377	-17.699	52	1.155e-23
10C	-0.9769	0.18583	-1.350	-0.604	-5.257	52	2.784e-06

So what's the use of `margeff`? The `contrast` and `lincon` functions can only handle *linear* functions of the model parameters. But in some cases the marginal effect is not a linear function of the model parameters. This is where the `margeff` function is useful.

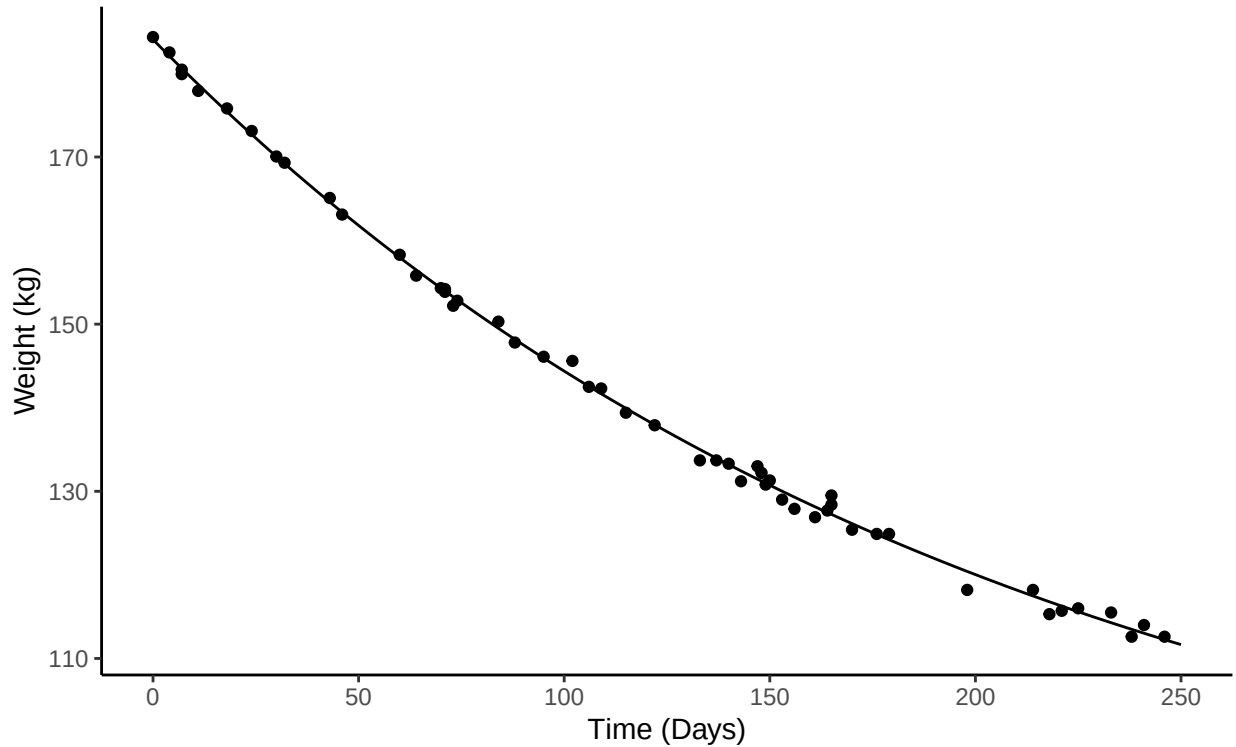
Example: Consider the following nonlinear model for the change in expected weight over time.

```
m <- nls(Weight ~ t1 + t2*2^(-Days/t3), data = MASS::wtloss,
  start = list(t1 = 90, t2 = 95, t3 = 120))
```

```
d <- data.frame(Days = seq(0, 250, by = 1))
d$yhat <- predict(m, newdata = d)
```

```
p <- ggplot(MASS::wtloss, aes(x = Days, y = Weight)) +
  geom_point() + theme_classic() +
  labs(y = "Weight (kg)", x = "Time (Days)") +
```

```
geom_line(aes(y = yhat), data = d)
plot(p)
```



The model is

$$E(Y) = \theta_1 + \theta_2 2^{-d/\theta_3},$$

where Y is weight and d is days. The discrete marginal effect of going from 50 to 100 days is

$$\underbrace{\theta_1 + \theta_2 2^{-100/\theta_3}}_{E(Y|d=100)} - \underbrace{(\theta_1 + \theta_2 2^{-50/\theta_3})}_{E(Y|d=50)} = \theta_2 (2^{-100/\theta_3} - 2^{-50/\theta_3}).$$

This is *not* a linear function of the model parameters, so we cannot use the usual methods like `contrast` or `lincon`. But we can make (approximate) inferences using the *delta method* (more on that later). The `margeff` function makes implementing this method relatively straight forward.

```
margeff(m, a = list(Days = 100), b = list(Days = 50))
```

```
estimate      se  lower  upper tvalue df    pvalue
-17.43 0.1292 -17.69 -17.17 -134.9 49 1.182e-64
```

```
margeff(m,
  a = list(Days = c(50,100,150,200)),
  b = list(Days = c(0,50,100,150)),
  cnames = c("0->50", "50->100", "100->150", "150->200"))
```

```
      estimate      se  lower  upper tvalue df    pvalue
0->50      -22.25 0.3291 -22.91 -21.59  -67.6 49 4.838e-50
50->100    -17.43 0.1292 -17.69 -17.17 -134.9 49 1.182e-64
100->150   -13.65 0.1033 -13.86 -13.44 -132.2 49 3.109e-64
150->200   -10.69 0.1606 -11.02 -10.37  -66.6 49 1.001e-49
```

Example: Consider the following model for the insecticide data.

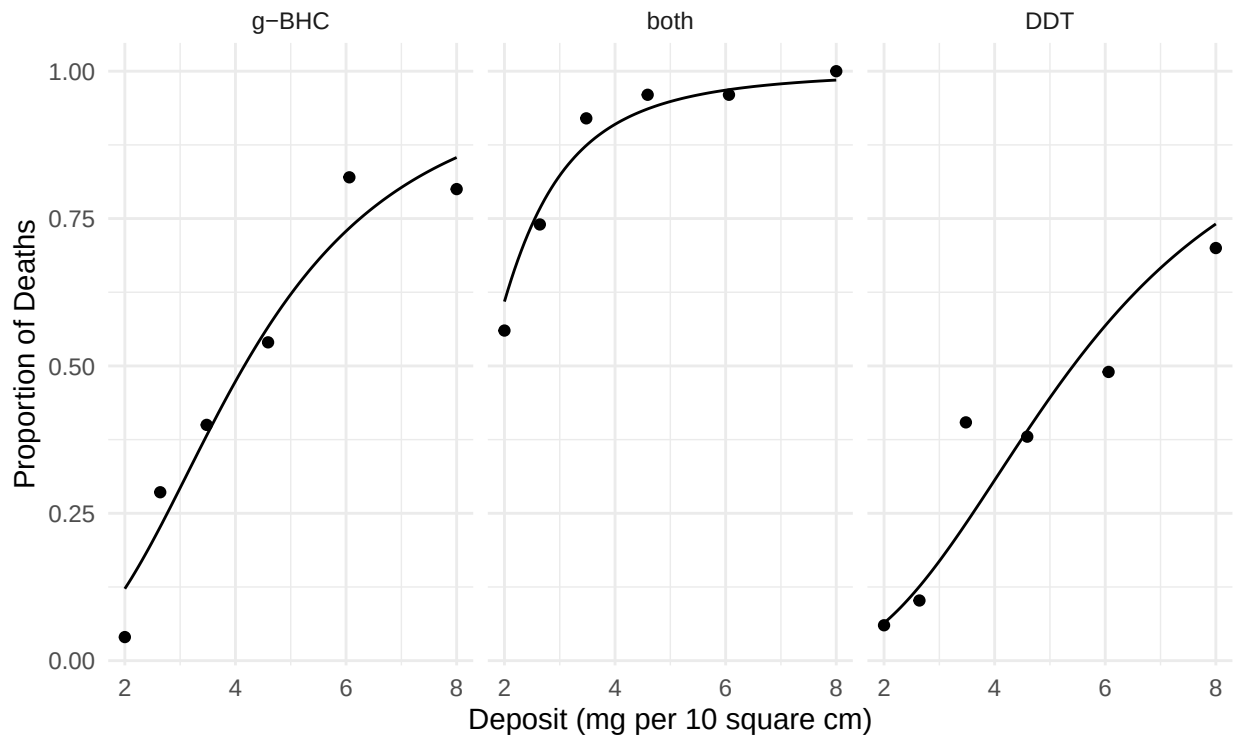
```

m <- glm(cbind(deaths, total-deaths) ~ log(deposit) + insecticide,
        family = binomial, data = insecticide)

d <- expand.grid(deposit = seq(2, 8, length = 100),
               insecticide = unique(insecticide$insecticide))
d$phat <- predict(m, newdata = d, type = "response")

p <- ggplot(insecticide, aes(x = deposit, y = deaths/total)) +
  geom_point() + facet_wrap(~ insecticide) +
  geom_line(aes(y = phat), data = d) + theme_minimal() +
  labs(x = "Deposit (mg per 10 square cm)",
       y = "Proportion of Deaths")
plot(p)

```



We know how to interpret the effects using *odds ratios*. Here are the odds ratios for the effect of doubling the deposit from 2 to 4 units.

```

contrast(m, tf = exp,
  a = list(deposit = 4, insecticide = c("g-BHC", "both", "DDT")),
  b = list(deposit = 2, insecticide = c("g-BHC", "both", "DDT")),
  cnames = c("g-BHC", "both", "DDT"))

```

	estimate	lower	upper
g-BHC	6.479	4.833	8.685
both	6.479	4.833	8.685
DDT	6.479	4.833	8.685

And here are the odds ratios for the effect of insecticide (g-BHC versus DDT).

```

contrast(m, tf = exp,
  a = list(deposit = c(2,4,6,8), insecticide = "g-BHC"),
  b = list(deposit = c(2,4,6,8), insecticide = "DDT"),

```

```
cnames = c("2", "4", "6", "8"))
```

	estimate	lower	upper
2	2.04	1.383	3.007
4	2.04	1.383	3.007
6	2.04	1.383	3.007
8	2.04	1.383	3.007

But with odds ratios we have to interpret effects in terms of *odds*. What if we want to interpret the effect on the *probability*? The discrete marginal effect is in terms of the *expected response* (here the expected proportion or, equivalently, the probability of death).

```
margeff(m,
  a = list(deposit = 4, insecticide = c("g-BHC", "both", "DDT")),
  b = list(deposit = 2, insecticide = c("g-BHC", "both", "DDT")),
  cnames = c("g-BHC", "both", "DDT"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
g-BHC	0.3517	0.02470	0.3033	0.4001	14.240	Inf	5.183e-46
both	0.3007	0.03650	0.2292	0.3723	8.239	Inf	1.737e-16
DDT	0.2424	0.02188	0.1995	0.2853	11.076	Inf	1.633e-28

Here are some discrete marginal effects of insecticide (g-BHC versus DDT).

```
margeff(m,
  a = list(deposit = c(2,4,6,8), insecticide = "g-BHC"),
  b = list(deposit = c(2,4,6,8), insecticide = "DDT"),
  cnames = c("2", "4", "6", "8"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
2	0.05821	0.01773	0.02346	0.09295	3.283	Inf	0.0010257
4	0.16753	0.04565	0.07806	0.25700	3.670	Inf	0.0002425
6	0.16034	0.04395	0.07420	0.24647	3.648	Inf	0.0002638
8	0.11275	0.03225	0.04955	0.17596	3.496	Inf	0.0004717

The appeal of the marginal effect here is that for many people probabilities are more intuitive than odds.

Example: Consider the following model for data from a study of the effect of blood plasma concentration/dilution on clotting time.

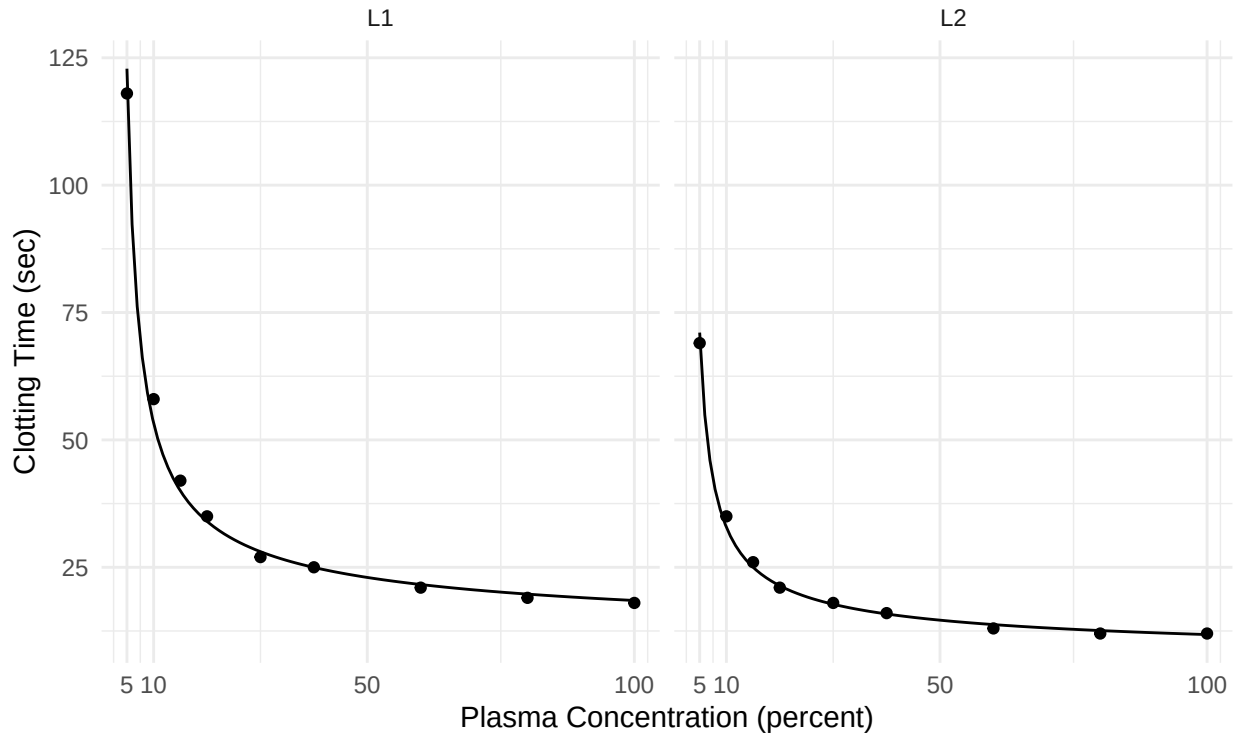
```
clotting <- data.frame(
  conc = rep(c(5,10,15,20,30,40,60,80,100), 2),
  time = c(118,58,42,35,27,25,21,19,18,69,35,26,21,18,16,13,12,12),
  lot = rep(c("L1", "L2"), each = 9)
)
head(clotting)
```

	conc	time	lot
1	5	118	L1
2	10	58	L1
3	15	42	L1
4	20	35	L1
5	30	27	L1
6	40	25	L1

```
m <- glm(time ~ lot + log(conc) + lot:log(conc),
  family = Gamma(link = inverse), data = clotting)
```

```
d <- expand.grid(conc = seq(5, 100, length = 100), lot = c("L1", "L2"))
d$yhat <- predict(m, newdata = d, type = "response")

p <- ggplot(clotting, aes(x = conc, y = time)) + theme_minimal() +
  geom_point() + facet_wrap(~ lot) + facet_wrap(~ lot) +
  labs(x = "Plasma Concentration (percent)", y = "Clotting Time (sec)") +
  scale_x_continuous(breaks = c(5, 10, 50, 100)) +
  geom_line(aes(y = yhat), data = d)
plot(p)
```



```
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.016554	0.0008655	-19.127	1.967e-11
lotL2	-0.007354	0.0016780	-4.383	6.252e-04
log(conc)	0.015343	0.0003872	39.626	8.851e-16
lotL2:log(conc)	0.008256	0.0007353	11.228	2.184e-08

This generalized linear model can be written as

$$\frac{1}{E(T_i)} = \beta_0 + \beta_1 l_i + \beta_2 \log_2 c_i + \beta_3 l_i \log_2 c_i,$$

or, equivalently,

$$E(T_i) = \frac{1}{\beta_0 + \beta_1 l_i + \beta_2 \log_2 c_i + \beta_3 l_i \log_2 c_i},$$

where T_i is clotting time, c_i is plasma concentration, and l_i is an indicator variable such that $l_i = 1$ if the i -th observation is from the second lot, and $l_i = 0$ otherwise.

Marginal effects of increasing the plasma concentration from 5 to 10 in each lot.

```
margeff(m,
  a = list(conc = 10, lot = c("L1", "L2")),
  b = list(conc = 5, lot = c("L1", "L2")),
  cnames = c("L1,5->10", "L2,5->10"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
L1,5->10	-69.6	4.808	-79.91	-59.28	-14.47	14	8.149e-10
L2,5->10	-38.2	2.711	-44.01	-32.38	-14.09	14	1.160e-09

Marginal effects of increasing from 5 to 10, 10 to 50, and 50 to 100 in the first lot.

```
margeff(m,
  a = list(conc = c(10,50,100), lot = "L1"),
  b = list(conc = c(5,10,50), lot = "L1"),
  cnames = c("L1,5->10", "L1,10->50", "L1,50->100"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
L1,5->10	-69.595	4.80805	-79.907	-59.283	-14.47	14	8.149e-10
L1,10->50	-30.259	0.71242	-31.787	-28.731	-42.47	14	3.376e-16
L1,50->100	-4.522	0.06961	-4.671	-4.373	-64.96	14	9.064e-19

Marginal effects for plasma concentration for the *second* lot.

```
margeff(m,
  a = list(conc = c(10,50,100), lot = "L2"),
  b = list(conc = c(5,10,50), lot = "L2"),
  cnames = c("L2,5->10", "L2,10->50", "L2,50->100"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
L2,5->10	-38.197	2.7107	-44.010	-32.383	-14.09	14	1.160e-09
L2,10->50	-18.244	0.4595	-19.230	-17.259	-39.71	14	8.606e-16
L2,50->100	-2.821	0.0436	-2.914	-2.727	-64.69	14	9.613e-19

Marginal effects for lot at three plasma concentrations.

```
margeff(m,
  a = list(conc = c(25,50,75), lot = c("L1")),
  b = list(conc = c(25,50,75), lot = c("L2")),
  cnames = c("25", "50", "75"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
25	11.246	0.5809	10.000	12.492	19.36	14	1.672e-11
50	8.388	0.4810	7.356	9.420	17.44	14	6.835e-11
75	7.301	0.4394	6.359	8.244	16.62	14	1.304e-10

“Instantaneous” Marginal Effects

Consider a regression model with (without loss of generality) two explanatory variables, X_1 and X_2 . Assuming that X_1 is *continuous*, the “instantaneous” marginal effect of X_1 at x_1 when $X_2 = x_2$ is

$$\lim_{\delta \rightarrow 0} \frac{E(Y|X_1 = x_1 + \delta, X_2 = x_2) - E(Y|X_1 = x_1, X_2 = x_2)}{\delta}.$$

This can also be written as

$$\left. \frac{\partial E(Y|X_1 = z, X_2 = x_2)}{\partial z} \right|_{z=x_1}$$

i.e., the partial derivative of $E(Y|X_1 = x_1, X_2 = x_2)$ with respect to and evaluated at x_1 .

Intuitively, this is the rate of change in the expected response at a specific value of the explanatory variable — i.e., the slope of the function at a specific point.

To compute this marginal effect we can either find the partial derivative analytically or approximate it numerically using

$$\frac{E(Y|X_1 = x_1 + \delta, X_2 = x_2) - E(Y|X_1 = x_1, X_2 = x_2)}{\delta}$$

where δ set to a small value relative to x_1 (this is called *numerical differentiation*).

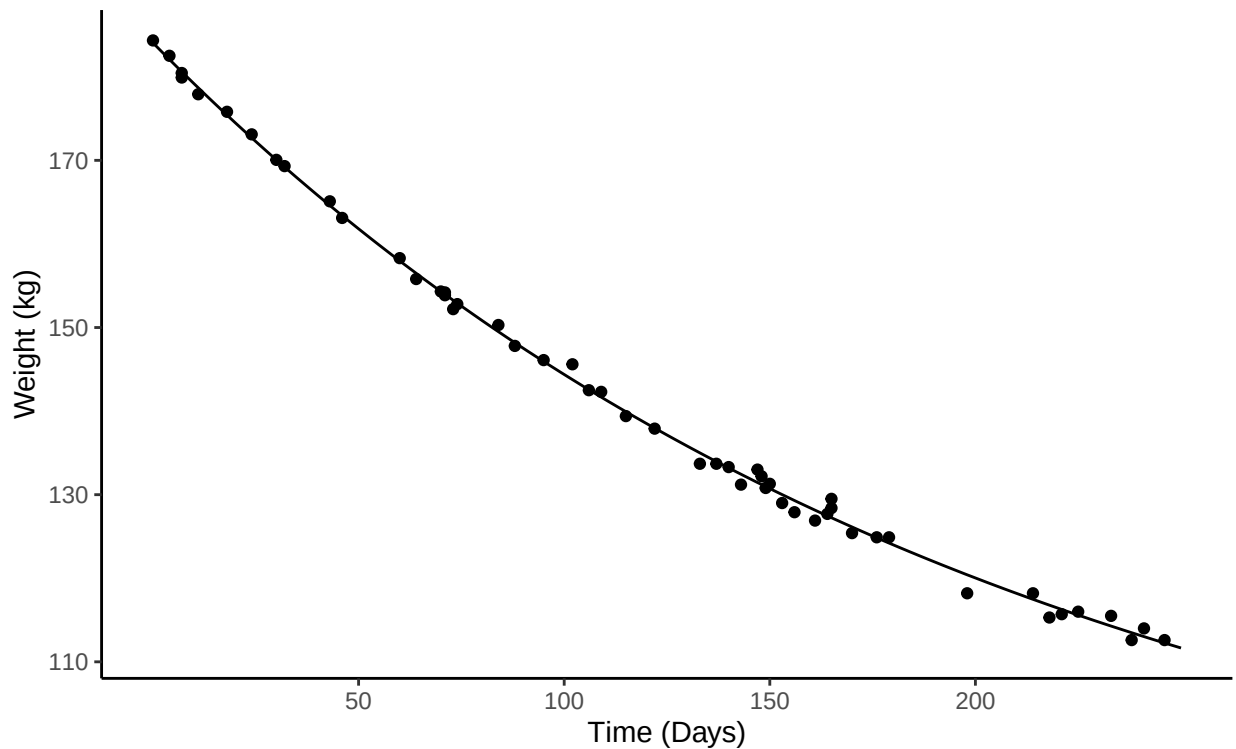
Note that instantaneous marginal effects are only defined for *continuous quantitative variables*.

Example: Consider again the nonlinear regression model for expected weight as a function of days.

```
m <- nls(Weight ~ t1 + t2*2^(-Days/t3), data = MASS::wtloss,
  start = list(t1 = 90, t2 = 95, t3 = 120))

d <- data.frame(Days = seq(0, 250, by = 1))
d$yhat <- predict(m, newdata = d)

p <- ggplot(MASS::wtloss, aes(x = Days, y = Weight)) +
  geom_point() + theme_classic() +
  labs(y = "Weight (kg)", x = "Time (Days)") +
  geom_line(aes(y = yhat), data = d) +
  scale_x_continuous(breaks = c(50, 100, 150, 200))
plot(p)
```



We can estimate the instantaneous marginal effects at 50, 100, 150, and 200 days.

```
margeff(m, delta = 0.001,
  a = list(Days = c(50, 100, 150, 200) + 0.001),
  b = list(Days = c(50, 100, 150, 200)),
  cnames = c("@50", "@100", "@150", "@200"))
```


	estimate	se	lower	upper	tvalue	df	pvalue
@50	-0.3929	0.004173	-0.4013	-0.3845	-94.14	49	4.924e-57
@100	-0.3077	0.001832	-0.3114	-0.3041	-168.03	49	2.529e-69
@150	-0.2411	0.002685	-0.2465	-0.2357	-89.79	49	4.936e-56
@200	-0.1888	0.003675	-0.1962	-0.1814	-51.39	49	2.759e-44

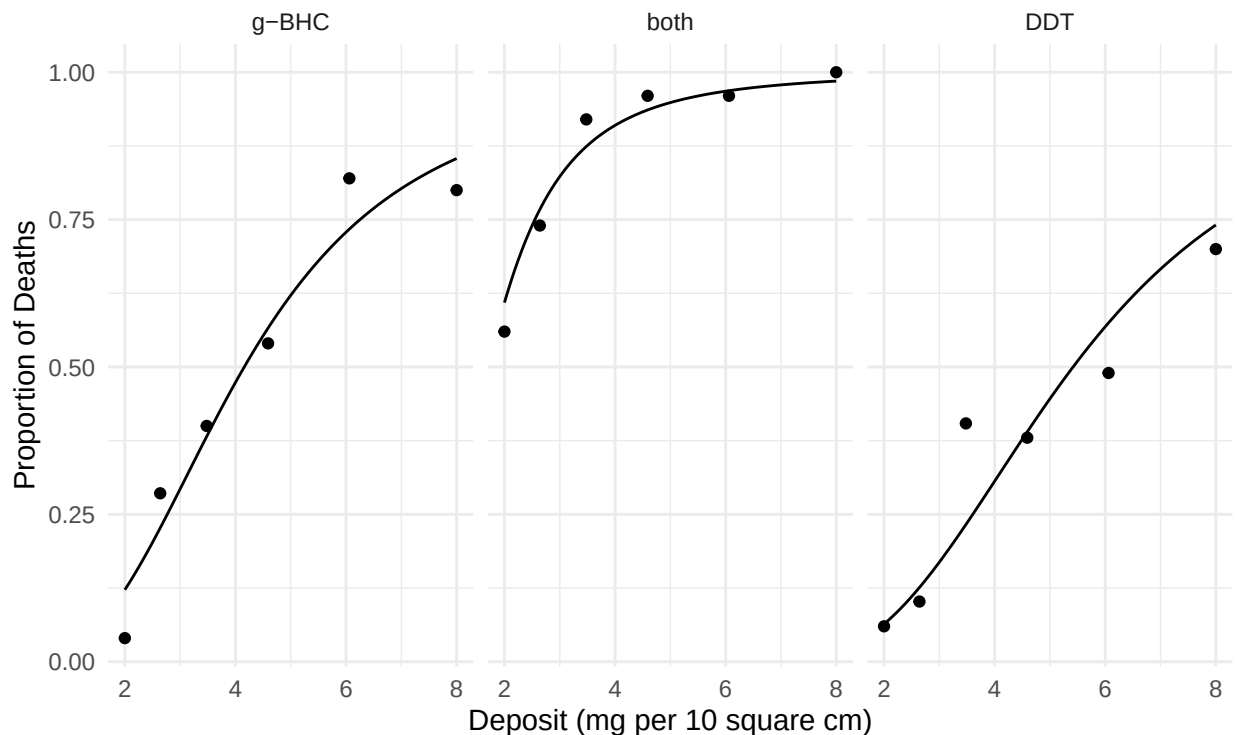
Note: To estimate an instantaneous marginal effect, add a relatively small value of δ to the `a` variable, and also specify this amount to the `delta` argument.

Example: Consider again the model for the insecticide data.

```
m <- glm(cbind(deaths, total-deaths) ~ log(deposit)
+ insecticide, family = binomial, data = insecticide)

d <- expand.grid(deposit = seq(2, 8, length = 100),
insecticide = unique(insecticide$insecticide))
d$phat <- predict(m, newdata = d, type = "response")

p <- ggplot(insecticide, aes(x = deposit, y = deaths/total)) +
geom_point() + facet_wrap(~ insecticide) +
geom_line(aes(y = phat), data = d) + theme_minimal() +
labs(x = "Deposit (mg per 10 square cm)", y = "Proportion of Deaths")
plot(p)
```

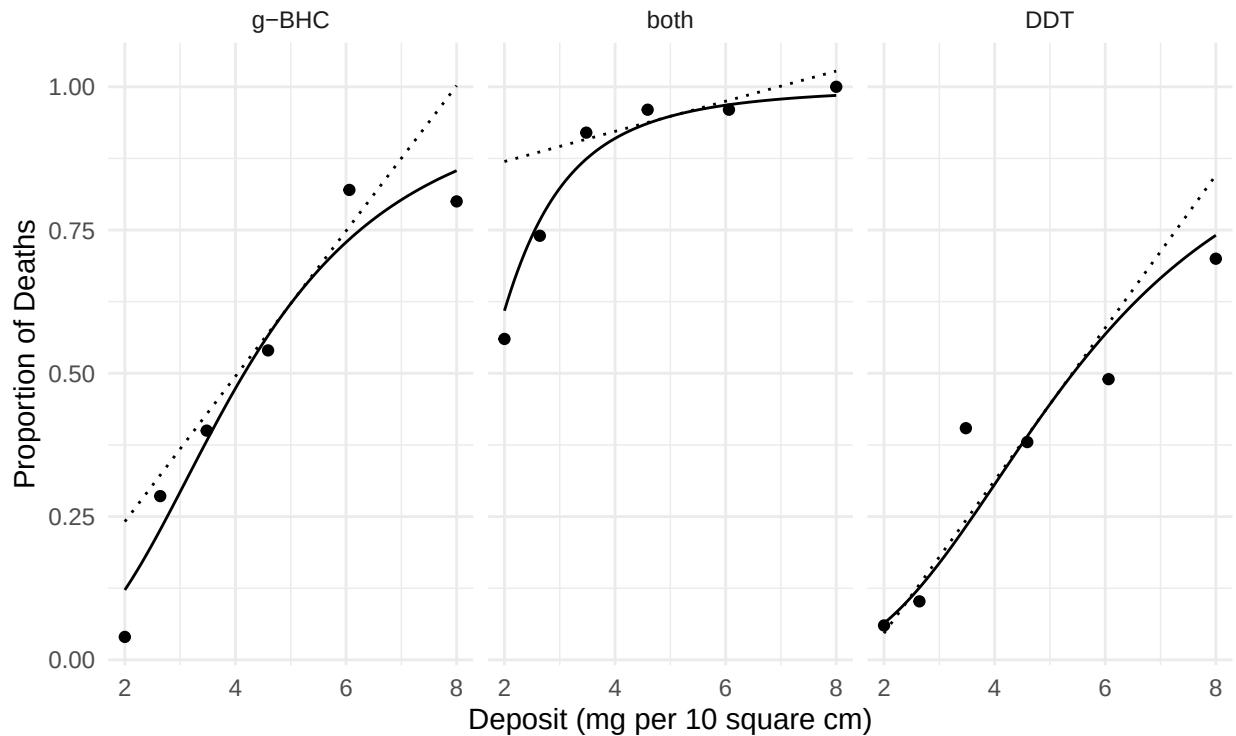


We can estimate the instantaneous marginal effect of deposit at a given amount of deposit, say 5 mg per 10 square cm.

```
margeff(m, delta = 0.001,
a = list(deposit = 5 + 0.001, insecticide = c("g-BHC", "both", "DDT")),
b = list(deposit = 5, insecticide = c("g-BHC", "both", "DDT")),
cnames = c("g-BHC", "both", "DDT"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
--	----------	----	-------	-------	--------	----	--------

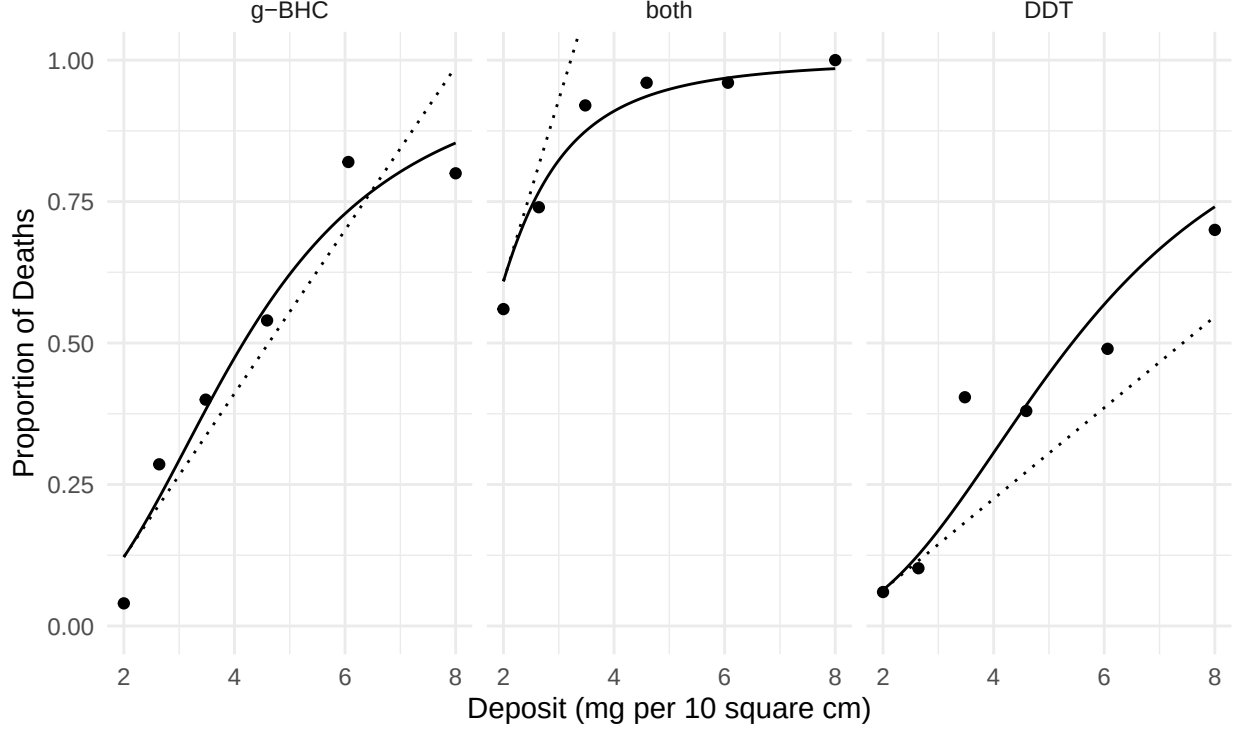
g-BHC	0.12680	0.009750	0.10769	0.1459	13.005	Inf	1.149e-38
both	0.02631	0.004281	0.01792	0.0347	6.146	Inf	7.941e-10
DDT	0.13321	0.011056	0.11154	0.1549	12.048	Inf	1.978e-33



Note that the instantaneous effect of deposit *depends on the deposit* because the probability is not a linear function of deposit.

```
margeff(m, delta = 0.001,
  a = list(deposit = 2 + 0.001, insecticide = c("g-BHC", "both", "DDT")),
  b = list(deposit = 2, insecticide = c("g-BHC", "both", "DDT")),
  cnames = c("g-BHC", "both", "DDT"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
g-BHC	0.14439	0.01575	0.11352	0.1753	9.168	Inf	4.839e-20
both	0.32078	0.03323	0.25565	0.3859	9.654	Inf	4.750e-22
DDT	0.08049	0.01180	0.05737	0.1036	6.824	Inf	8.878e-12



Instantaneous Marginal Effects for Generalized Linear Models

Recall that in a GLM that $E(Y) = g^{-1}(\eta)$ where $\eta = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$. Consider a GLM where $\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$. The instantaneous marginal effect of X_1 at x_1 is

$$\frac{\partial E(Y|X_1 = x_1, X_2 = x_2)}{\partial x_1} = \frac{\partial g^{-1}(\eta)}{\partial x_1} = \frac{\partial g^{-1}(\eta)}{\partial \eta} \frac{\partial \eta}{\partial x_1} = \frac{\partial g^{-1}(\eta)}{\partial \eta} \beta_1$$

by the “chain rule” for (partial) derivatives.

Suppose that $E(Y) = e^\eta$ (i.e., log link function) where $\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. Then

$$\frac{\partial g^{-1}(\eta)}{\partial \eta} \beta_1 = \frac{\partial e^\eta}{\partial \eta} \beta_1 = e^\eta \beta_1 = E(Y) \beta_1.$$

Suppose now that $E(Y) = e^\eta / (1 + e^\eta)$ (i.e., logit link function). Then

$$\frac{\partial g^{-1}(\eta)}{\partial \eta} \beta_1 = \frac{\partial e^\eta / (1 + e^\eta)}{\partial \eta} \beta_1 = \frac{e^\eta}{(1 + e^\eta)^2} \beta_1 = E(Y)[1 - E(Y)] \beta_1.$$

Suppose now that $E(Y) = \eta$ (e.g., identity link function). Then

$$\frac{\partial g^{-1}(\eta)}{\partial \eta} \beta_1 = \frac{\partial \eta}{\partial \eta} \beta_1 = \beta_1.$$

Things get a little more complicated if X_1 is a *transformed* explanatory variable or represents an interaction.

Suppose $E(Y) = \beta_0 + \beta_1 \log(x_1) + \beta_2 x_2$. Then

$$\frac{\partial g^{-1}(\eta)}{\partial \eta} \frac{\partial \eta}{\partial x_1} = \frac{\partial \eta}{\partial x_1} = \beta_1 / x_1.$$

Suppose $E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2$. Then

$$\frac{\partial g^{-1}(\eta)}{\partial \eta} \frac{\partial \eta}{\partial x_1} = \frac{\partial \eta}{\partial x_1} = \beta_1 + 2\beta_2 x_1.$$

Suppose $E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$. Then

$$\frac{\partial g^{-1}(\eta)}{\partial \eta} \frac{\partial \eta}{\partial x_1} = \frac{\partial \eta}{\partial x_1} = \beta_1 + \beta_3 x_2.$$

Fortunately, `margeff` does the calculus!

Discrete Marginal Effects as Percent Change

Consider a regression model with (without loss of generality) two explanatory variables, X_1 and X_2 . The *percent change* in the expected response when changing X_1 from x_b to x_a when $X_2 = x_2$ is

$$\frac{E(Y|X_1 = x_a, X_2 = x_2) - E(Y|X_1 = x_b, X_2 = x_2)}{E(Y|X_1 = x_b, X_2 = x_2)} \times 100\%.$$

or, equivalently,

$$\left[\frac{E(Y|X_1 = x_a, X_2 = x_2)}{E(Y|X_1 = x_b, X_2 = x_2)} - 1 \right] \times 100\%.$$

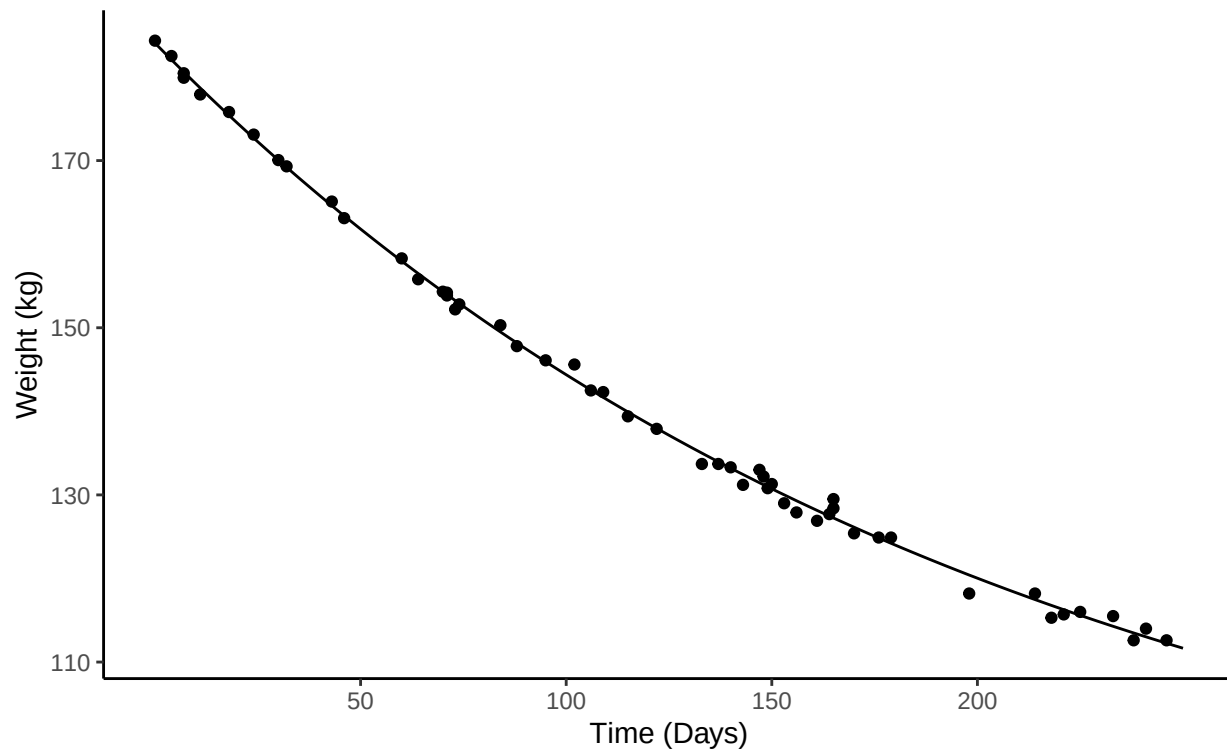
Note that the *sign* indicates if it is a percent increase or decrease.

Example: Consider again the weight loss model.

```
m <- nls(Weight ~ t1 + t2*2^(-Days/t3), data = MASS::wtloss,
  start = list(t1 = 90, t2 = 95, t3 = 120))

d <- data.frame(Days = seq(0, 250, by = 1))
d$yhat <- predict(m, newdata = d)

p <- ggplot(MASS::wtloss, aes(x = Days, y = Weight)) +
  geom_point() + theme_classic() +
  labs(y = "Weight (kg)", x = "Time (Days)") +
  geom_line(aes(y = yhat), data = d) +
  scale_x_continuous(breaks = c(50,100,150,200))
plot(p)
```



Consider the percent change in expected weight from 50 to 100 days. This is

$$\frac{\theta_1 + \theta_2 2^{-100/\theta_3} - \theta_1 - \theta_2 2^{-50/\theta_3}}{\theta_1 + \theta_2 2^{-50/\theta_3}} = \frac{\theta_2 2^{-100/\theta_3} - \theta_2 2^{-50/\theta_3}}{\theta_1 + \theta_2 2^{-50/\theta_3}}.$$

We can estimate the percent change in expected weight from 50 to 100 days as follows.

```
margeff(m, a = list(Days = 100), b = list(Days = 50), type = "percent")
```

```
estimate      se  lower  upper tvalue df    pvalue
-10.77 0.07673 -10.93 -10.62 -140.4 49 1.666e-65
```

We can do it for several 50 day increments as well.

```
margeff(m, type = "percent",
  a = list(Days = c(50,100,150,200)),
  b = list(Days = c(0,50,100,150)),
  cnames = c("0->50", "50->100", "100->150", "150->200"))
```

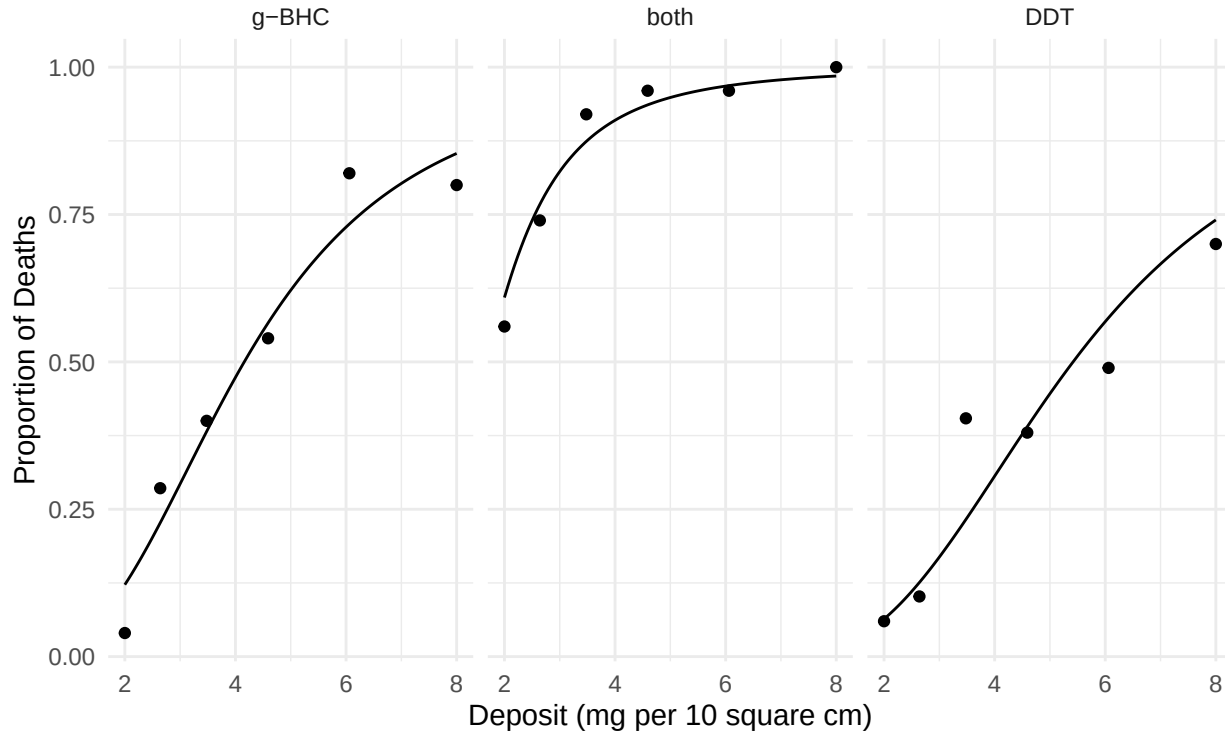
```
      estimate      se  lower  upper  tvalue df    pvalue
0->50    -12.089 0.15795 -12.406 -11.771  -76.53 49 1.171e-52
50->100  -10.771 0.07673 -10.926 -10.617 -140.37 49 1.666e-65
100->150  -9.456 0.06627  -9.589  -9.323 -142.69 49 7.488e-66
150->200  -8.180 0.12091  -8.423  -7.937  -67.66 49 4.660e-50
```

Example: Consider again the model for the insecticide data.

```
m <- glm(cbind(deaths, total-deaths) ~ log(deposit) + insecticide,
  family = binomial, data = insecticide)

d <- expand.grid(deposit = seq(2, 8, length = 100),
  insecticide = levels(insecticide$insecticide))
d$phat <- predict(m, newdata = d, type = "response")
```

```
p <- ggplot(insecticide, aes(x = deposit, y = deaths/total)) +
  geom_point() + facet_wrap(~ insecticide) +
  geom_line(aes(y = phat), data = d) + theme_minimal() +
  labs(x = "Deposit (mg per 10 square cm)", y = "Proportion of Deaths")
plot(p)
```



We can estimate the percent change in the probability of death from 4 to 6 mg per 10 square cm.

```
margeff(m, type = "percent",
  a = list(deposit = 6, insecticide = c("g-BHC", "both", "DDT")),
  b = list(deposit = 4, insecticide = c("g-BHC", "both", "DDT")),
  cnames = c("g-BHC", "both", "DDT"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
g-BHC	53.821	6.567	40.950	66.692	8.196	Inf	2.488e-16
both	6.372	1.111	4.195	8.548	5.738	Inf	9.604e-09
DDT	85.621	11.031	63.999	107.242	7.761	Inf	8.394e-15

Note that here the percent change depends on where we make the increment.

```
margeff(m, type = "percent",
  a = list(deposit = 8, insecticide = c("g-BHC", "both", "DDT")),
  b = list(deposit = 6, insecticide = c("g-BHC", "both", "DDT")),
  cnames = c("g-BHC", "both", "DDT"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
g-BHC	17.153	1.9971	13.238	21.067	8.589	Inf	8.798e-18
both	1.764	0.3625	1.053	2.474	4.866	Inf	1.140e-06
DDT	30.364	3.2940	23.908	36.820	9.218	Inf	3.021e-20

We can also estimate the percent change in the probability of death between two insecticides.

```

margeff(m, type = "percent",
  a = list(deposit = c(2,4,6,8), insecticide = "g-BHC"),
  b = list(deposit = c(2,4,6,8), insecticide = "DDT"),
  cnames = c("2", "4", "6", "8"))

```

	estimate	se	lower	upper	tvalue	df	pvalue
2	91.29	34.752	23.173	159.40	2.627	Inf	0.008620
4	54.72	19.037	17.406	92.03	2.874	Inf	0.004049
6	28.21	9.132	10.315	46.11	3.090	Inf	0.002005
8	15.22	4.904	5.607	24.83	3.103	Inf	0.001914

Discrete Marginal Effects as Multiplicative Factors

Consider a regression model with (without loss of generality) two explanatory variables, X_1 and X_2 . A multiplicative factor to describe the effect of changing X_1 from x_b to x_a when $X_2 = x_2$ is

$$f = \frac{E(Y|X_1 = x_a, X_2 = x_2)}{E(Y|X_1 = x_b, X_2 = x_2)},$$

meaning that

$$E(Y|X_1 = x_a, X_2 = x_2) = f \times E(Y|X_1 = x_b, X_2 = x_2).$$

Example: Consider again the model for the insecticide data.

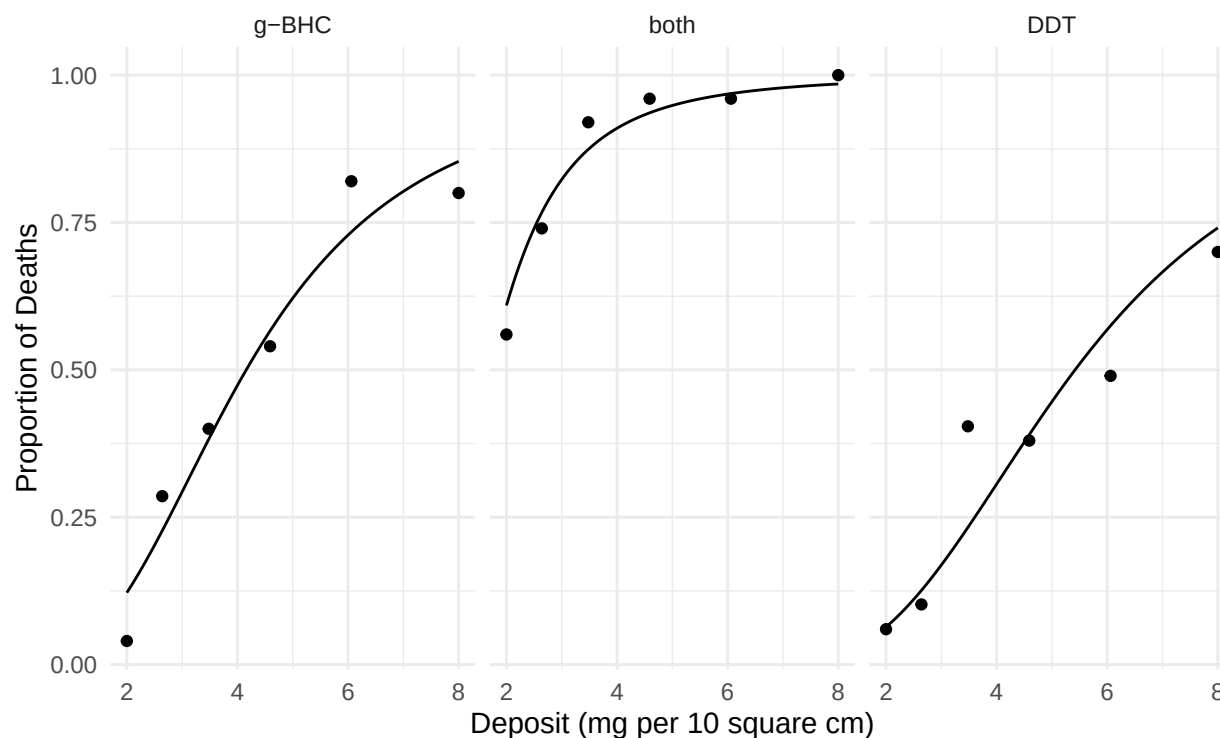
```

m <- glm(cbind(deaths, total-deaths) ~ log(deposit) + insecticide,
  family = binomial, data = insecticide)

d <- expand.grid(deposit = seq(2, 8, length = 100),
  insecticide = levels(insecticide$insecticide))
d$phat <- predict(m, newdata = d, type = "response")

p <- ggplot(insecticide, aes(x = deposit, y = deaths/total)) +
  geom_point() + facet_wrap(~ insecticide) +
  geom_line(aes(y = phat), data = d) + theme_minimal() +
  labs(x = "Deposit (mg per 10 square cm)", y = "Proportion of Deaths")
plot(p)

```



We can estimate the factor by which we increase probability by increasing deposit from 4 to 6 mg per 10 square cm.

```
margeff(m, type = "factor",
  a = list(deposit = 6, insecticide = c("g-BHC", "both", "DDT")),
  b = list(deposit = 4, insecticide = c("g-BHC", "both", "DDT")),
  cnames = c("g-BHC", "both", "DDT"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
g-BHC	1.538	0.06567	1.410	1.667	23.42	Inf	2.436e-121
both	1.064	0.01111	1.042	1.085	95.78	Inf	0.000e+00
DDT	1.856	0.11031	1.640	2.072	16.83	Inf	1.562e-63

We can also estimate the factor for comparing both insecticides with g-BHC only.

```
margeff(m, type = "factor",
  a = list(deposit = c(2,4,6,8), insecticide = "both"),
  b = list(deposit = c(2,4,6,8), insecticide = "g-BHC"),
  cnames = c("2", "4", "6", "8"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
2	4.994	0.87192	3.285	6.703	5.728	Inf	1.016e-08
4	1.921	0.14204	1.642	2.199	13.523	Inf	1.139e-41
6	1.328	0.05464	1.221	1.435	24.312	Inf	1.469e-130
8	1.154	0.03098	1.093	1.215	37.242	Inf	1.444e-303

Using Different Kinds of Marginal Effects

Marginal effects give us a variety of ways to summarize the statistical relationship between a response variable and an explanatory variable.

Example: Consider the following model for the `ToothGrowth` data.

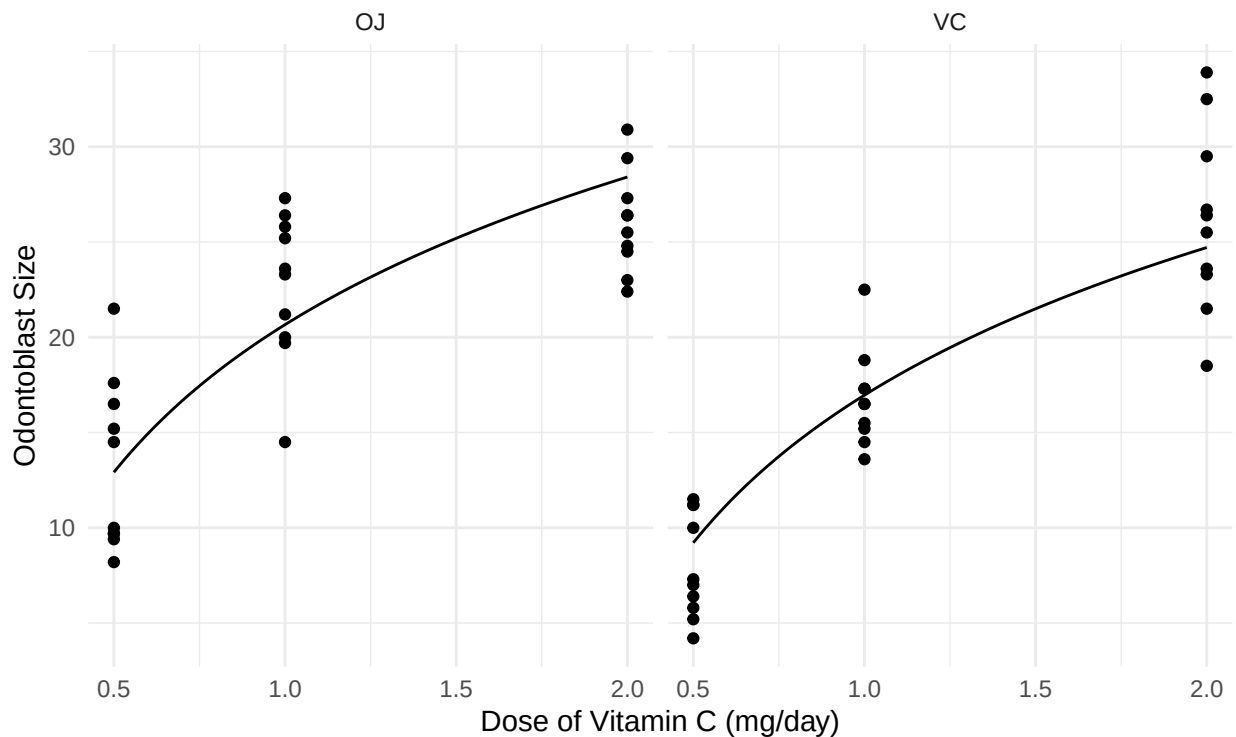

```

m <- lm(len ~ log(dose) + supp, data = ToothGrowth)

d <- expand.grid(dose = seq(0.5, 2, length = 100), supp = c("OJ", "VC"))
d$yhat <- predict(m, d)

p <- ggplot(ToothGrowth, aes(x = dose, y = len)) +
  geom_point() + facet_wrap(~ supp) +
  geom_line(aes(y = yhat), data = d) +
  labs(x = "Dose of Vitamin C (mg/day)", y = "Odontoblast Size") +
  theme_minimal()
plot(p)

```



We can use discrete marginal effects, such as when increasing dose from 0.5 to 1 mg/day.

```

margeff(m, cnames = c("OJ", "VC"),
  a = list(dose = 1.0, supp = c("OJ", "VC")),
  b = list(dose = 0.5, supp = c("OJ", "VC")))

```

	estimate	se	lower	upper	tvalue	df	pvalue
OJ	7.748	0.6091	6.528	8.967	12.72	57	2.736e-18
VC	7.748	0.6091	6.528	8.967	12.72	57	2.736e-18

We can use instantaneous effects, such as the instantaneous effect at 1 mg/day.

```

margeff(m, cnames = c("OJ", "VC"), delta = 0.001,
  a = list(dose = 1 + 0.001, supp = c("OJ", "VC")),
  b = list(dose = 1, supp = c("OJ", "VC")))

```

	estimate	se	lower	upper	tvalue	df	pvalue
OJ	11.17	0.8783	9.413	12.93	12.72	57	2.736e-18
VC	11.17	0.8783	9.413	12.93	12.72	57	2.736e-18

We can use the percent change, such as when increasing dose from 0.5 to 1 mg/day.

```
margeff(m, cnames = c("OJ", "VC"), type = "percent",
  a = list(dose = 1.0, supp = c("OJ", "VC")),
  b = list(dose = 0.5, supp = c("OJ", "VC")))
```

	estimate	se	lower	upper	tvalue	df	pvalue
OJ	59.98	8.222	43.52	76.45	7.296	57	1.022e-09
VC	84.07	13.754	56.53	111.61	6.112	57	9.411e-08

We can use a multiplicative factor, such as when increasing dose from 0.5 to 1 mg/day.

```
margeff(m, cnames = c("OJ", "VC"), type = "factor",
  a = list(dose = 1.0, supp = c("OJ", "VC")),
  b = list(dose = 0.5, supp = c("OJ", "VC")))
```

	estimate	se	lower	upper	tvalue	df	pvalue
OJ	1.600	0.08222	1.435	1.764	19.46	57	7.556e-27
VC	1.841	0.13754	1.565	2.116	13.38	57	3.081e-19

Note: There are functions in other packages for estimating some kinds of marginal effects (e.g., see the package **marginaleffects**).